Overview

Small, repeatedly flooded municipalities with older building stock have no engineering tool to evaluate whether community-scale drainage investment or distributed building hardening better reduces flood damage. PIs Napolitano (structural vulnerability and community co-design), Wauthier (SAR processing and uncertainty quantification), and Busse (hydrological modeling and mass balance), all at Penn State, propose a closed-loop cyber-physical system (CPS) that couples the physical behavior of pre-code load-bearing masonry and timber, absent from existing fragility frameworks, with a computational core that turns free national data and community memory into building-level damage estimates. It senses through USGS gauges and NOAA precipitation operationally, with commercial ICEYE SAR for one-time calibration and open Sentinel-1 for flood-extent context; its computational core is a Bayesian framework fusing a GIS-parameterizable mass balance model (inputs from free national data: USGS 3DEP, NLCD, SSURGO, NHD StreamStats) with community-elicited duration priors and a gauge-stage-to-terrain depth projection; and the community closes the loop through a co-developed two-mode interface for scenario-based planning, forecast-driven staging, and post-event model updating. Human decisions are the actuators at every timescale.

The CPS is developed and validated in Montpelier, Vermont across two flood events (2023 calibration, 2024 held-out validation), with 271 downtown pre-code masonry buildings each carrying ICEYE-measured duration, depth, and field-recorded damage state. **RQ1:** To what extent can this CPS produce building-level damage assessments decision-adequate for retrofit allocation without hydraulic simulation? **RQ2:** Under what conditions does community knowledge, entered as Bayesian priors, improve assessments relative to a GIS-only baseline, and where does it function as substitute versus complement?

Intellectual Merit

The central contribution is the closed loop these two questions form, a working CPS in which free national data and community knowledge together support real investment decisions with no hydraulic simulation. Realizing it required three advances. First, a *computational* advance: a Bayesian framework that infers building-level depth and duration from sparse, interval-censored observations, fuses them with a GIS-parameterizable mass balance model and community-elicited priors, and propagates uncertainty to a decision-adequate damage estimate. Second, a *physical* advance: duration-dependent fragility surfaces for pre-code load-bearing masonry, absent from existing frameworks; the CPS senses flood signatures and informs interventions whose effects reenter future flood behavior. Third, treating locally held *community knowledge* as a structural calibration input, characterizing when Bayesian priors improve accuracy over a GIS-only baseline and where they function as substitutes versus complements. Together these establish that free national data fused with community knowledge suffices for the investment decisions small municipalities face.

Broader Impacts

The framework gives small municipalities probabilistic damage assessments and pre-storm decision support that does not currently exist. A new community needs only a USGS gauge ID, national GIS layer access, and one workshop; no commercial SAR, hydrologist, or hydraulic model is required. The CIR co-development methodology, documented as a transferable protocol, converts tacit process knowledge into quantitative model inputs measured against a geospatial baseline. One postdoctoral researcher, co-advised by all three PIs across the project's technical lanes, receives four years of cross-disciplinary training spanning SAR processing, mass balance modeling, probabilistic structural assessment, and community-engaged research. All code will be released under an MIT license.